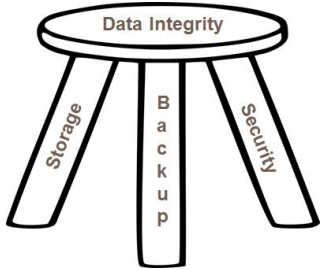
Big Data to Knowledge(BD2K)

# Best Practices for Biomedical Research Data Management

## Module 4: Data Storage and Security Information Security

Brian Coleman, CISSP, CISM, UMass Medical School




This video will discuss breaches in security for healthcare data, causes of breaches, and some ways to prevent breaches of various kinds. Topics of importance include phishing, ransomware, malware and network-based backup options.

CC BY NC SA

# New England healthcare organizations reporting large information security breaches in last 5 years

*…40+ others in our region*
*…Over 1200 nationwide*

*Source: media reports and OCR reporting*
*https://ocrportal.hhs.gov/ocr/breach/breach_report.jsf*

Hartford Hospital
Connect to healthier.™

SEACOAST RADIOLOGY

Lahey Hospital & Medical Center

Massachusetts Eye and Ear

UCONN HEALTH

Women & Infants

North Country Hospital
Where caring runs deep

Maine Medical Center
MaineHealth
centered around you

Beth Israel Deaconess Medical Center

BRIGHAM AND WOMEN'S HOSPITAL

Boston Children's Hospital
Until every child is well

SPEARE Memorial Hospital

LANDMARK MEDICAL CENTER

TUFTS Health Plan

South Shore Hospital

St. Elizabeth's Medical Center

UMassMemorial Medical Center

PARTNERS HEALTHCARE

BRUINS WIN IN OVERTIME, 3-2, PUSH RED WINGS TO THE BRINK — C1

The Boston Globe

FRIDAY, APRIL 25, 2014

## In the news

### Late shift

Friday: Turning rainy at night; high 58-63, low 41-46

Saturday: Rainy, cooler; high 47-52, low 39-44

High tide: 8:30 a.m., 9:04 p.m.

Sunrise: 5:48 Sunset: 7:37

Complete report, B13

# Cyberattack hits Children's Hospital

## May be the work of group opposing teen's treatment

By Michael B. Farrell and Patricia Wen
GLOBE STAFF

The infamous computer hacker network known as Anonymous threatened to attack Boston Children's Hospital over the child custody case involving Justina Pelletier last month, just a few weeks before the medical center's website was subjected to numerous cyber-assaults.

The anti-authority members of Anonymous sometimes appear in Guy Fawkes masks.

Anonymous has made its interest in the case clear. Several weeks ago, the group claimed responsibility for an attack on the website of Wayside Youth and Family Support Network, the Framingham residential facility where 15-year old Justina has been living since January under state custody.

After the more recent attack on Children's, some patients and medical personnel could not use their online accounts to check appointments, test results, and other case information after the hospital shut down those Web pages.

The threats from Anonymous are the latest to emerge against

# Firefigl deal wo raise pa by 18.8

City's 6-year p put at $92.4m is expected ne

By Meghan E. I

Boston Children's Hospital

HARVARD MEDICAL SCHOOL TEACHING HOSPITAL

# Mass Eye and Ear Breach

- Physician's laptop stolen and was **not** encrypted.
- Laptop contained Protected Health Information for ~3500 patients

## Risks to patient information and exposure of organization to regulatory response increased due to

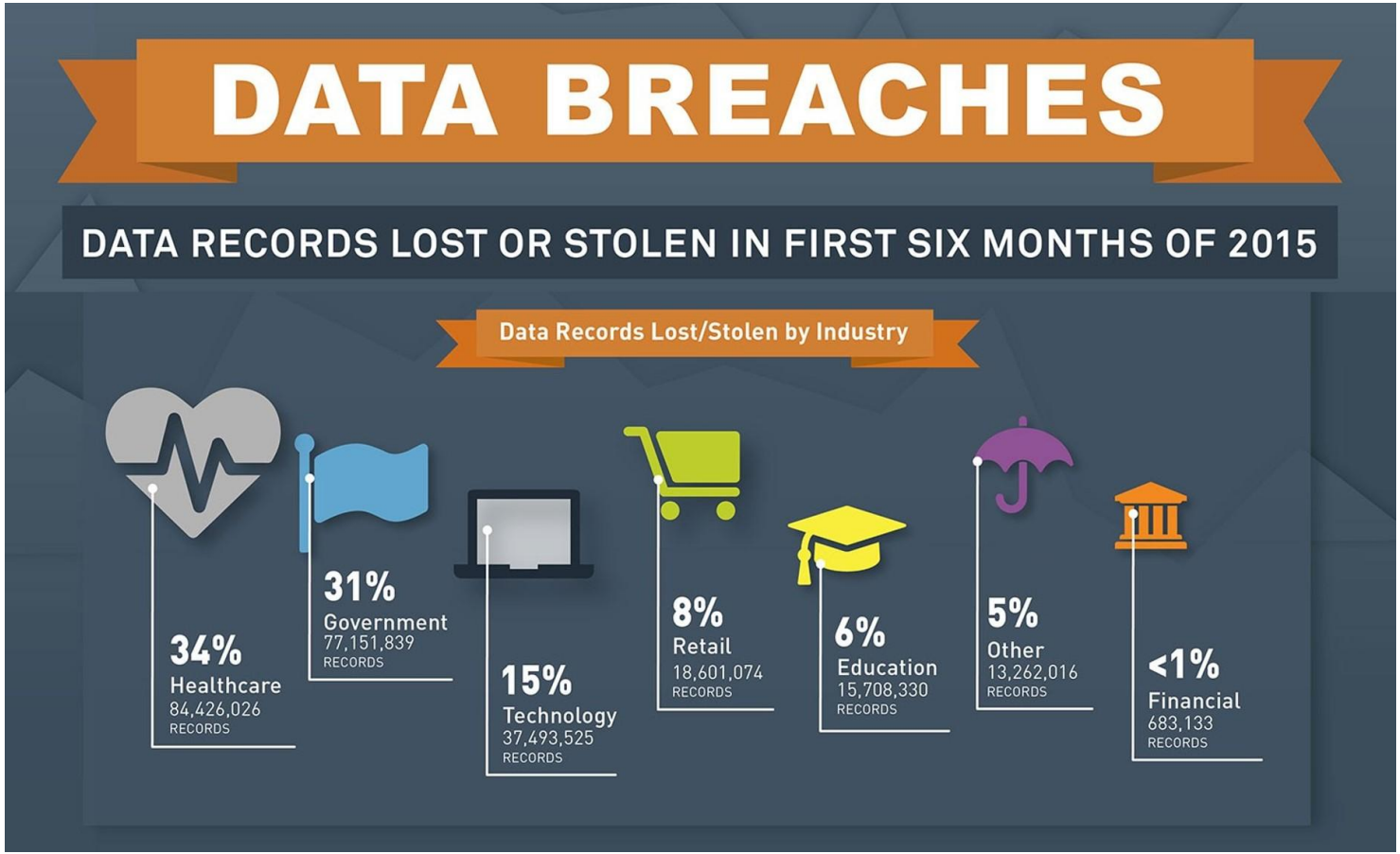
- Delay in physician reporting theft to IT and Privacy Office
- Delay in activating Laptop Lo-jack product to wipe laptop contents

**Responsibility may be broader than organization would expect. BYOD, doesn't matter, you still must protected patient data and follow best practices**

# The Internet of Things (IofT)

**470,000 Vehicles At Risk After Hackers "Take Control & Crash" Jeep Cherokee From A Sofa 10 Miles Away…**

**The on-board Internet connection is a 'super nice vulnerability' for hackers**


DATA BREACHES
DATA RECORDS LOST OR STOLEN IN FIRST SIX MONTHS OF 2015
Data Records Lost/Stolen by Industry
34%
Healthcare
84,426,026
RECORDS
31%
Government
77,151,839
RECORDS
15%
Technology
37,493,525
RECORDS
8%
Retail
18,601,074
RECORDS
6%
Education
15,708,330
RECORDS
5%
Other
13,262,016
RECORDS
<1%
Financial
683,133
RECORDS

# Breach Root Cause: All Industries

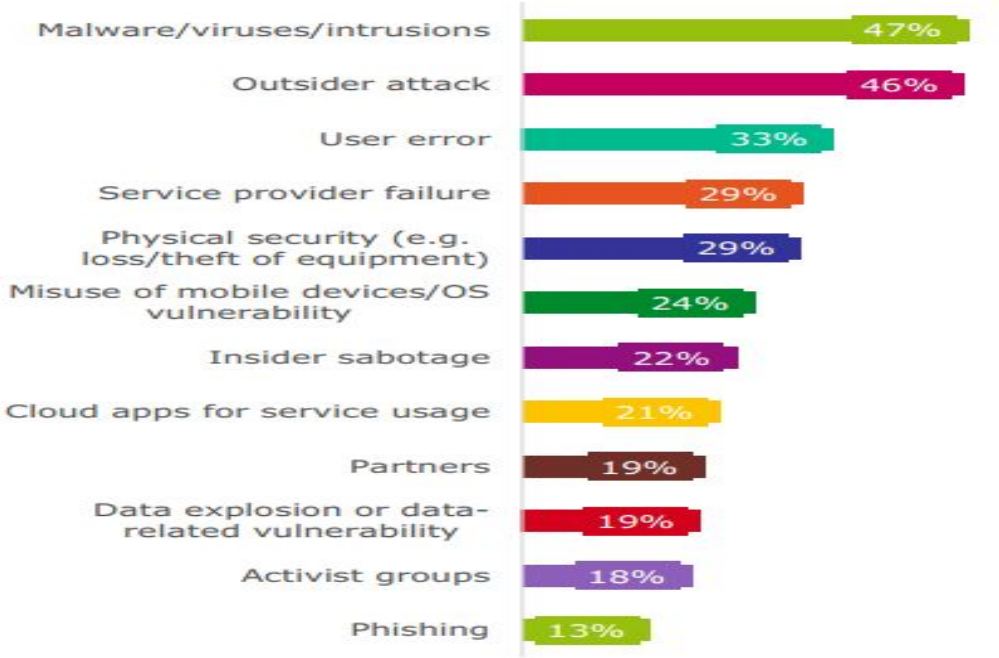



**Figure 16:** "What was the root cause of the security breaches your organization experienced?", *only asked to respondents that have experienced a security breach (1241 respondents)*

# Focus: Healthcare

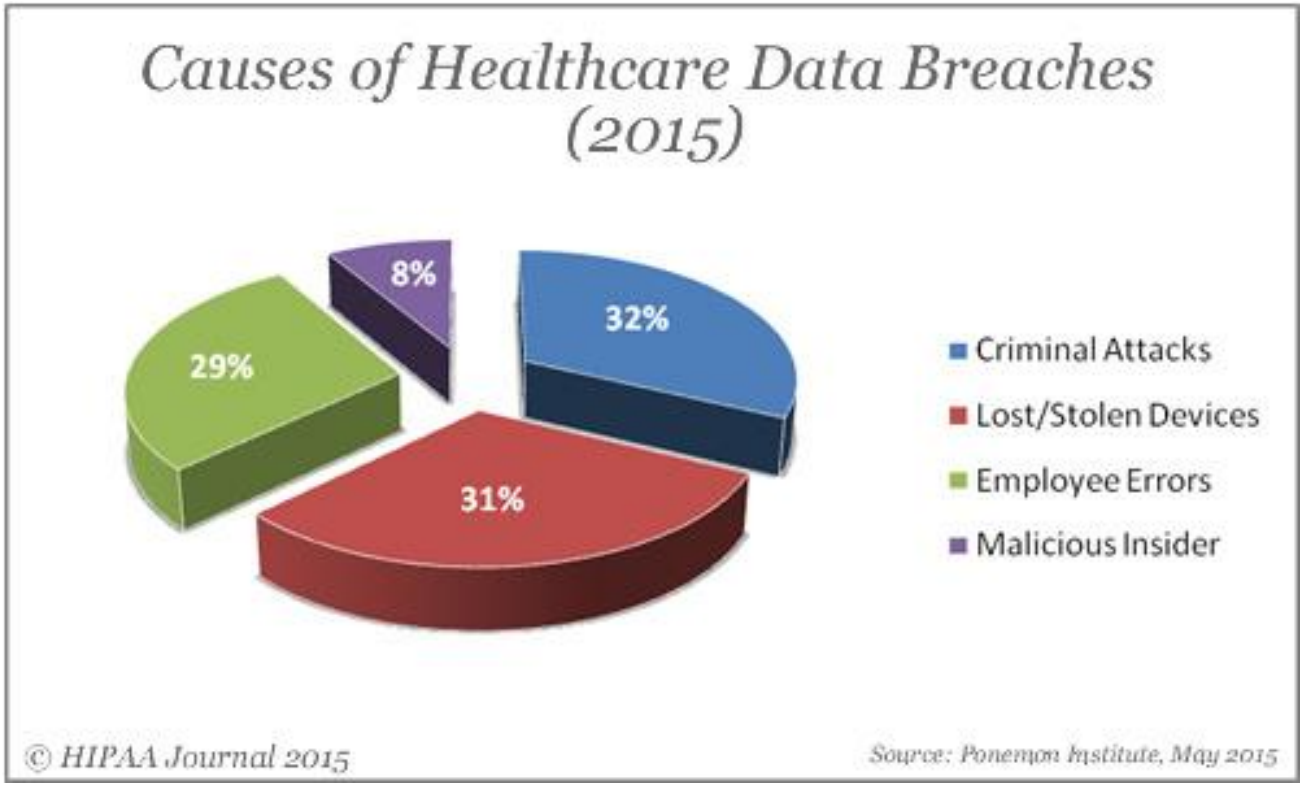
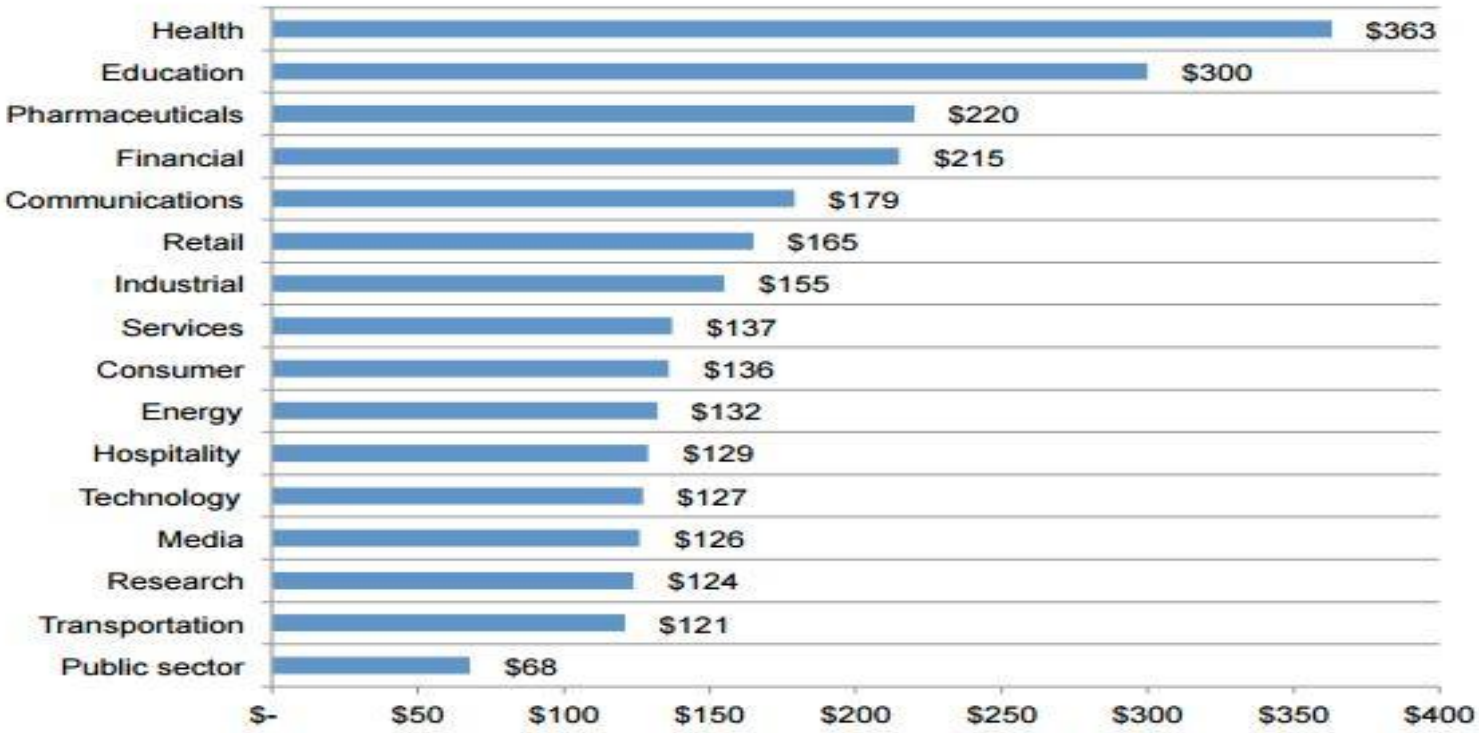


**Figure 4. Per capita cost by industry classification**
Consolidated view (n=350), measured in US$



# Phishing: Don’t Get Hooked

Send Money, Pay Online

paypal.com-us.cgi-bin-webscr-cmd.login-submit-dispatch.74fghghs68g484iky4mn86we8r46d4h38df4

BOGUS URL

Sign Up | Log In | Help | Security and Protection

PayPal

Home | Personal | Business | Developers

Get to Know PayPal | Pay Online | Send Money | Get Paid | Products & Services

Account login

Email address

PayPal password

Go to

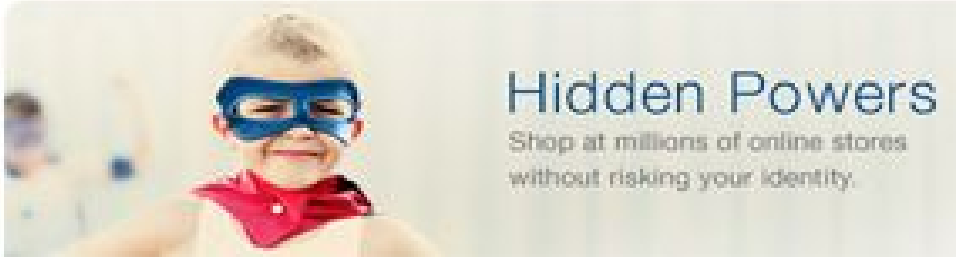
My account

Log In

Problem with login?

WELCOME TO PayPal

The world's most loved way to pay and get paid. Learn More



yahooads-offer.tk - Mozilla Firefox

File Edit View History Bookmarks Tools Help Related Links

Notice no HTTPS and the random domain extentsion ".tk"

http://yahooads-offer.tk/

Google Search Bookmarks AutoLink AutoFill Sign in

http://en-us.start2.m...zilla:en-US:official | yahooads-offer.tk

YAHOO! MAIL

Yahoo! - Help

**Get the complete picture. Sign up for Yahoo! Mail.**

Enjoy tons of features and fun, simple ways to share photos. Sign up now -- it's FREE!

Tour PhotoMail

To access Yahoo! Mail... you need a Yahoo! ID.

**Don't have a Yahoo! ID?**
Signing up is easy.

Sign up for Yahoo!

**Already have a Yahoo! ID?**
Sign in.

Yahoo! ID:

Password:

Remember my ID on this computer

Sign In

MODE: Standard | Secure
Forget your ID or password?
Sign-in help

Get mobile. Get messages.
Yahoo! alerts you of new email and lets you read them on your mobile phone.

AntiVirus that works twice as hard.
Your Yahoo! Mail scans and cleans email attachments to help keep nasty viruses out of your life.

Wanna share in our trophy?
*PC Magazine* knows a thing or two about free email, including which one stands out. Show me

**One Yahoo! ID. So much fun!**
Use your single ID for everything from checking Mail to checking out Yahoo! Music, Photos, Messenger, and more.

Copyright © 2005 Yahoo! Inc. All rights reserved. Copyright/IP Policy | Terms of Service | Guide to Online

Done

start | yahooads-offer.tk - ...

# The Hover Method

billing and credit card information.

To keep your account active, please follow these steps by August 29, 2005.

1. Go to http://smallbusiness.yahoo.com/accountupdate.
2. Enter your domain name. If you have multiple domain names that need to be reactivated, you will need to g
3. Sign in with the Yahoo! ID associated with your account.

http://218.246.224.203/yahoo/accountupdate/

# Would You Click on This?

**From:** dcemops [mailto:gshs1@westnet.com.au]
**Sent:** Saturday, June 06, 2015 11:08 PM
**To:** XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
**Subject:** Re:6/7/2015 3:08:04 PM

http://twojzlotnik.pl/qkgns/xoocmczywwiwqfptzpdyzpwh.gvywfdfwlsyckobwzqlkfsrzqdxztgobusf

gshs1

# It Could Be Phishing, If:

- The Email/Message Comes From An Unrecognized Sender
- If they are asking you to confirm personal or financial information over the internet and/or make urgent requests for the information
- If the email is not personalized (and is asking for info or contains attachments)
- Try to elicit an emotional response out of you by threatening you with frightening consequences to get you to respond quickly

# What to do if you receive a suspicious email:

- Never click on any links or open attachments with emails that you are not expecting or do not know the sender
- Always simply delete the email if it looks suspicious
- Never respond to “Pop Up” windows asking for information
- Never provide personal or financial information on an untrusted web site
- Call the Help Desk if you suspect that you have fallen victim to a Phishing email

# Ransomware

Your personal files are encrypted!
Your important files encryption produced on this computer: photos, videos, documents, etc. Here is a complete list of encrypted files, and you can personally verify this.
Encryption was produced using a unique public key RSA-2048 generated for this computer. To decrypt files you need to obtain the private key.
The single copy of the private key, which will allow you to decrypt the files, located on a secret server on the Internet; the server will destroy the key after a time specified in this window. After that, nobody and never will be able to restore files...
To obtain the private key for this computer, which will automatically decrypt files, you need to pay 100 USD / 100 EUR / similar amount in another currency.
Click «Next» to select the method of payment and the currency.
Any attempt to remove or damage this software will lead to the immediate destruction of the private key by server.
Private key will be destroyed on
9/8/2013
5:52 PM
Time left
56 : 16 : 12
Next >>